NAMAN KUMAR

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SUMMARY

Computer Science undergraduate aiming for a career in data science and AI. Built real-time computer vision, machine learning and full-stack web applications, plus embedded prototypes. Proficient in Python, C and JavaScript. Winner of a college hackathon and an ideathon.

ACHIEVEMENTS

- **Winner, Zero to One Hackathon** | Dayananda Sagar College of Engineering
- **Winner, Ideathon** | Department of Physics, Dayananda Sagar College of Engineering

EDUCATION

B.E. in Computer Science, Dayananda Sagar College of Engineering, Bengaluru *Expected 2029*
CGPA: 8.75 | Coursework: Data Structures & Algorithms, Computer Organization & Architecture

TECHNICAL SKILLS

Languages: Python, C++, C, C/C++, HTML, CSS
Hardware & Embedded: Arduino Uno, I2C, PWM, ultrasonic sensors, servo motors, LCD

PROJECTS

ETHICare VitalSense: Wearable Stress and Cardiac Signal Analysis *Codethon project*

- Built a full-stack platform that detects stress levels and cardiac abnormalities from wearable physiological data (WESAD and ECG datasets).
- Designed a two-service architecture: a Node.js service for authentication, experiments and reports, and a Python FastAPI microservice for preprocessing, feature extraction and ML inference.
- Created animated signal visualizations and compared model results against published literature.

Signspeak: Real-Time Sign Language Detection *Personal project*

- Built a browser-based app that recognizes the ASL alphabet, numbers and common words from a live camera feed and converts them to text.
- Implemented gesture recognition using finger extension and curvature analysis with multi-frame stabilization to reduce false detections.
- Added voice output for accessibility, a gesture library and a detection history dashboard; all processing runs in the browser, so no video leaves the device.

ThoughtFirewall: AI Manipulation Detection Dashboard *Hackathon project*

- Designed a real-time dashboard that scores manipulation risk (0-100) in ads, clickbait and misinformation and explains the tactics in plain language.
- Streamed live analysis events over WebSockets and visualized a six-agent reasoning pipeline, with a reframe mode showing alternative interpretations and source-trust analytics.

Smart Mess Attendance System *Personal project*

- Built a web app to cut food wastage in a college mess: students mark meal attendance (eat/skip) and staff see live headcount statistics.
- Developed REST endpoints (/api/attendance, /api/stats) with form validation and toast notifications.

Arduino Motion Detection and Alarm System (SEAMS) *College project*

- Programmed an Arduino Uno system that detects motion via ultrasonic sensing, shows distance and status on an I2C LCD, and escalates through LED alert levels.
- Triggered a servo and buzzer alarm after a no-motion timeout; added separate Demo and Real modes with different timing thresholds.